1 Gain law

(bars denote normalized quantities: lengths by R; $a_o := \Delta t/(\gamma_N R)$ [1/pN]; $\bar{a} := a/a_{\text{nom}}$; $\bar{f} := a_o f$, so that $\bar{a}\,\bar{f} = \Delta\bar{w}$; $g := \bar{w} - \bar{w}_s = \bar{h}$, $\bar{w}$ = particle centre along the wall normal, $\bar{w}_s$ = position of the wall surface on the same axis (plane: 0; cell: the visible surface), known, contact at $g = 1$; c_0, c_1, d_1, d_2 = the [2/2] coefficients, constants of the wall, fitted on the band $1.1 \le g \le 5.5$ and used there only; $e_\theta = \theta - \hat{\theta}$)

$$N(g) := g^2 + c_1\, g + c_0, \qquad N'(g) = 2g + c_1$$
$$D(g) := g^2 + d_1\, g + d_2, \qquad D'(g) = 2g + d_1$$
$$\bar{a}(g;\, c_0, c_1, d_1, d_2) = \frac{N(g)}{D(g)}$$
$$1 - \bar{a} = \frac{(d_1 - c_1)\, g + (d_2 - c_0)}{D(g)}$$
$$\bar{a}'(g) := \frac{\partial \bar{a}}{\partial \bar{w}} = \frac{N'(g) - \bar{a}\, D'(g)}{D(g)}$$
$$\bar{a}''(g) := \frac{\partial^2 \bar{a}}{\partial \bar{w}^2} = \frac{2\,(1 - \bar{a}) - 2\,\bar{a}'\, D'(g)}{D(g)}$$

$$\Big(\frac{\partial \bar{a}}{\partial c_0}\Big)_g = \frac{1}{D(g)}, \qquad \Big(\frac{\partial \bar{a}}{\partial c_1}\Big)_g = \frac{g}{D(g)}, \qquad \Big(\frac{\partial \bar{a}}{\partial d_1}\Big)_g = -\frac{\bar{a}\, g}{D(g)}, \qquad \Big(\frac{\partial \bar{a}}{\partial d_2}\Big)_g = -\frac{\bar{a}}{D(g)}$$
$$\Big(\frac{\partial \bar{a}'}{\partial c_0}\Big)_g = -\frac{D'(g)}{D(g)^2}, \qquad \Big(\frac{\partial \bar{a}'}{\partial c_1}\Big)_g = \frac{D(g) - g\, D'(g)}{D(g)^2}$$
$$\Big(\frac{\partial \bar{a}'}{\partial d_1}\Big)_g = \frac{\bar{a}\, g\, D'(g)}{D(g)^2} - \frac{\bar{a}}{D(g)} - \frac{\bar{a}'\, g}{D(g)}, \qquad \Big(\frac{\partial \bar{a}'}{\partial d_2}\Big)_g = \frac{\bar{a}\, D'(g)}{D(g)^2} - \frac{\bar{a}'}{D(g)}$$
$$a_w = a_o\, \bar{a}_w$$

$$\bar{a}(1) = \frac{1 + c_1 + c_0}{1 + d_1 + d_2}, \qquad \bar{a}'(1) = \frac{(2 + c_1)(1 + d_1 + d_2) - (1 + c_1 + c_0)(2 + d_1)}{(1 + d_1 + d_2)^2}, \qquad \lim_{g \to \infty} g\,(1 - \bar{a}) = d_1 - c_1$$

2 True dynamics

$\big(\delta\bar{w}_j[k] := \delta\bar{w}[k - d + j - 1],\ j = 1, \ldots, d + 1,\ d = 2\big)$

$$
\begin{aligned}
\bar{w}[k] &= \bar{w}_d[k] - \delta\bar{w}_3[k] \\
\bar{w}_d[k+1] &= \bar{w}_d[k] + \Delta\bar{w}_d[k] \\
\bar{a}_w[k] &= \frac{N(\bar{w}[k] - \bar{w}_s)}{D(\bar{w}[k] - \bar{w}_s)} \\
\delta\bar{w}_1[k+1] &= \delta\bar{w}_2[k] \\
\delta\bar{w}_2[k+1] &= \delta\bar{w}_3[k] \\
\delta\bar{w}_3[k+1] &= \lambda_c\,\delta\bar{w}_3[k] - F_{dw}[k]\,e_{\bar{a}_w}[k] - \varepsilon_w[k] \\
\Delta\bar{w}[k] &= \Delta\bar{w}_d[k] + (1-\lambda_c)\,\delta\bar{w}_3[k] + F_{dw}[k]\,e_{\bar{a}_w}[k] + \varepsilon_w[k] \\
c_0[k+1] &= c_0[k] \\
c_1[k+1] &= c_1[k] \\
d_1[k+1] &= d_1[k] \\
d_2[k+1] &= d_2[k]
\end{aligned}
$$

$$
F_{dw}[k] = \bar{f}_{dw}[k] + (1-\lambda_c)\sum_{i=1}^{d} \bar{f}_{dw}[k-i]
$$

$$
\varepsilon_w[k] = \bar{a}_w[k]\,\bar{f}_T[k] + (1-\lambda_c)\sum_{i=1}^{d} \bar{a}_w[k-i]\,\bar{f}_T[k-i] - (1-\lambda_c)\,n_w[k]
$$

3 Estimator A: four constant states, gain computed (no a slot)

($\hat{g}[k] := \bar{w}_d[k] - \delta\hat{\bar{w}}_3[k] - \bar{w}_s$; $\hat{N} := N(\hat{g};\,\hat{c}_0, \hat{c}_1)$, $\hat{D} := D(\hat{g};\,\hat{d}_1, \hat{d}_2)$; controller slots: $5 = d_1$, $7 = c_0$, $8 = c_1$, $9 = d_2$)

$$
\mathbf{x}_A = \left[\, \delta\bar{w}_1,\ \delta\bar{w}_2,\ \delta\bar{w}_3,\ c_0,\ c_1,\ d_1,\ d_2 \,\right]^\top
$$

$$
\begin{aligned}
\hat{\bar{a}}_w[k] &= \frac{\hat{N}[k]}{\hat{D}[k]} \\
\delta\hat{\bar{w}}_1[k+1] &= \delta\hat{\bar{w}}_2[k] \\
\delta\hat{\bar{w}}_2[k+1] &= \delta\hat{\bar{w}}_3[k] \\
\delta\hat{\bar{w}}_3[k+1] &= \lambda_c\,\delta\hat{\bar{w}}_3[k] \\
\hat{c}_0[k+1] &= \hat{c}_0[k] \\
\hat{c}_1[k+1] &= \hat{c}_1[k] \\
\hat{d}_1[k+1] &= \hat{d}_1[k] \\
\hat{d}_2[k+1] &= \hat{d}_2[k]
\end{aligned}
$$

4 Error dynamics A

($\mathbf{e}_A = [\, e_{\delta\bar{w}_1},\ e_{\delta\bar{w}_2},\ e_{\delta\bar{w}_3},\ e_{c_0},\ e_{c_1},\ e_{d_1},\ e_{d_2} \,]^\top$, coefficients at the estimate)

$$e_{\bar{a}_w}[k] = -\,\bar{a}'[k]\,e_{\delta\bar{w}_3}[k] + \frac{1}{\hat{D}[k]}\,e_{c_0}[k] + \frac{\hat{g}[k]}{\hat{D}[k]}\,e_{c_1}[k] - \frac{\hat{\bar{a}}_w[k]\,\hat{g}[k]}{\hat{D}[k]}\,e_{d_1}[k] - \frac{\hat{\bar{a}}_w[k]}{\hat{D}[k]}\,e_{d_2}[k]$$

$$e_{\delta\bar{w}_1}[k+1] = e_{\delta\bar{w}_2}[k]$$
$$e_{\delta\bar{w}_2}[k+1] = e_{\delta\bar{w}_3}[k]$$
$$e_{\delta\bar{w}_3}[k+1] = \big(\lambda_c + F_{dw}[k]\,\bar{a}'[k]\big)e_{\delta\bar{w}_3}[k] - F_{dw}[k]\,\frac{1}{\hat{D}[k]}\,e_{c_0}[k] - F_{dw}[k]\,\frac{\hat{g}[k]}{\hat{D}[k]}\,e_{c_1}[k] + F_{dw}[k]\,\frac{\hat{\bar{a}}_w[k]\,\hat{g}[k]}{\hat{D}[k]}\,e_{d_1}[k] + F_{dw}[k]\,\frac{\hat{\bar{a}}_w[k]}{\hat{D}[k]}\,e_{d_2}[k] - \varepsilon_w[k]$$
$$e_{c_0}[k+1] = e_{c_0}[k]$$
$$e_{c_1}[k+1] = e_{c_1}[k]$$
$$e_{d_1}[k+1] = e_{d_1}[k]$$
$$e_{d_2}[k+1] = e_{d_2}[k]$$

$$F_e^A[k] = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & \lambda_c + F_{dw}[k]\,\bar{a}'[k] & -\frac{F_{dw}[k]}{\hat{D}[k]} & -\frac{F_{dw}[k]\,\hat{g}[k]}{\hat{D}[k]} & \frac{F_{dw}[k]\,\hat{\bar{a}}_w[k]\,\hat{g}[k]}{\hat{D}[k]} & \frac{F_{dw}[k]\,\hat{\bar{a}}_w[k]}{\hat{D}[k]} \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

5 Measurement

$\big(\nabla_d\bar{w}_d[k] := \bar{w}_d[k] - \bar{w}_d[k-d];\ \hat{g}_d[k] := \hat{g}[k] - \nabla_d\bar{w}_d[k];\ \hat{N}_d := N(\hat{g}_d;\,\hat{c}_0,\hat{c}_1),\ \hat{D}_d := D(\hat{g}_d;\,\hat{d}_1,\hat{d}_2)\big)$

$$y_1[k] = \delta\bar{w}_1[k] + n_w[k]$$
$$y_2[k] = \bar{a}_w[k-d] + n_a[k]$$
$$\hat{y}_1[k] = \delta\hat{\bar{w}}_1[k]$$
$$\hat{\bar{a}}_d[k] = \frac{\hat{N}_d[k]}{\hat{D}_d[k]}$$
$$\hat{y}_2[k] = \hat{\bar{a}}_d[k]$$

$$H^A[k] = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\bar{a}'(\hat{g}_d[k]) & \frac{1}{\hat{D}_d[k]} & \frac{\hat{g}_d[k]}{\hat{D}_d[k]} & -\frac{\hat{\bar{a}}_d[k]\,\hat{g}_d[k]}{\hat{D}_d[k]} & -\frac{\hat{\bar{a}}_d[k]}{\hat{D}_d[k]} \end{bmatrix}$$

$$\bar{a}'(\hat{g}_d) = \bar{a}'(\hat{g}) - \nabla_d\bar{w}_d\,\bar{a}''(\hat{g}) + \ldots, \qquad \Big(\frac{\partial\bar{a}}{\partial\theta}\Big)_{\hat{g}_d} = \Big(\frac{\partial\bar{a}}{\partial\theta}\Big)_{\hat{g}} - \nabla_d\bar{w}_d\,\Big(\frac{\partial\bar{a}'}{\partial\theta}\Big)_{\hat{g}} + \ldots$$

$$\mathbf{e}_y[k] = H^A[k]\,\mathbf{e}^-[k] + \begin{bmatrix} n_w[k] \\ n_a[k] \end{bmatrix}$$

6 Noise

$$Q^A[k] = Q_{33}[k]\,\mathbf{g}_A\,\mathbf{g}_A^\top, \qquad \mathbf{g}_A = \begin{bmatrix} 0,\ 0,\ -1,\ 0,\ 0,\ 0,\ 0 \end{bmatrix}^\top$$

$$Q_{33}[k] = \kappa_T\Big(\hat{\bar{a}}_w[k] + (1-\lambda_c)^2\big(\hat{\bar{a}}_w[k-1] + \hat{\bar{a}}_w[k-2]\big)\Big) + (1-\lambda_c)^2\sigma_{n_w}^2, \qquad \kappa_T = \frac{4k_BT\,a_o}{R}$$

$$R[k] = \mathrm{diag}\big(R_1,\ R_2[k]\big), \qquad R_1 = \frac{\sigma_n^2}{R^2}, \qquad R_2[k] = 2a_{\mathrm{cov}}^2\,IF_{\mathrm{var}}\big(\hat{\bar{a}}_w[k]\big)\big(\hat{\bar{a}}_w[k] + \xi\big)^2$$

7 Observability

$(\Phi(k,k_0) := F_e[k-1]\cdots F_e[k_0],\ \Phi(k_0,k_0)=I;\ N$ window length)

$(O_w := R_w^{-1/2}\,[\,H[k_0]\Phi(k_0,k_0);\ \ldots;\ H[k_0+N]\Phi(k_0+N,k_0)\,];\ O_w\,\mathbf{v}_i=\sigma_i\,\mathbf{u}_i,\ \sigma_1\ge\cdots\ge\sigma_7)$

$$\Delta\bar{w}_d[k]=0\ \ \forall k,\ \delta\hat{\bar{w}}_3\to 0 \quad\Longrightarrow\quad F_{dw}[k]\to 0$$
$$\nabla_d\bar{w}_d[k]=0,\ \hat{g}[k]=\hat{g}\text{ constant},\ \hat{\bar{a}}_w[k]=\hat{\bar{a}}_w\text{ constant}$$
$$H^A[k]_{2,4:7}\to\frac{1}{\hat{D}}\left[\,1,\ \hat{g},\ -\hat{\bar{a}}_w\,\hat{g},\ -\hat{\bar{a}}_w\,\right]\text{ constant in }k$$
$$\Longrightarrow\quad \left\{\mathbf{e}: e_{c_0}+\hat{g}\,e_{c_1}-\hat{\bar{a}}_w\,\hat{g}\,e_{d_1}-\hat{\bar{a}}_w\,e_{d_2}=0\right\}\text{ unobservable (dimension 3)}$$

heights $\bar{w}_1,\ldots,\bar{w}_4$:
$$\det\begin{bmatrix}1 & g_1 & -\bar{a}_1 g_1 & -\bar{a}_1\\ 1 & g_2 & -\bar{a}_2 g_2 & -\bar{a}_2\\ 1 & g_3 & -\bar{a}_3 g_3 & -\bar{a}_3\\ 1 & g_4 & -\bar{a}_4 g_4 & -\bar{a}_4\end{bmatrix}\neq 0$$

8 Fixed height

($\bar{w}_d[k]=\bar{w}_h$ for all k; $g_h := \bar{w}_h-\bar{w}_s$; $\bar{a}_h := \bar{a}(\bar{w}_h)$ the plant's gain there; $r=$ the one number a single height pins)

$$\frac{g_h^2+c_1 g_h+c_0}{g_h^2+d_1 g_h+d_2}=\bar{a}_h \quad\Longleftrightarrow\quad c_0+c_1\,g_h-\bar{a}_h\,d_2-\bar{a}_h\,g_h\,d_1=g_h^2\,(\bar{a}_h-1)$$

$$r(\hat{c}_0,\hat{c}_1,\hat{d}_1,\hat{d}_2;\,\bar{w}_h) := \hat{c}_0+\hat{c}_1\,g_h-\bar{a}_h\,\hat{d}_2-\bar{a}_h\,g_h\,\hat{d}_1-g_h^2\,(\bar{a}_h-1)$$

four heights:
$$\begin{bmatrix}1 & g_1 & -\bar{a}_1 g_1 & -\bar{a}_1\\ 1 & g_2 & -\bar{a}_2 g_2 & -\bar{a}_2\\ 1 & g_3 & -\bar{a}_3 g_3 & -\bar{a}_3\\ 1 & g_4 & -\bar{a}_4 g_4 & -\bar{a}_4\end{bmatrix}\begin{bmatrix}c_0\\ c_1\\ d_1\\ d_2\end{bmatrix}=\begin{bmatrix}g_1^2(\bar{a}_1-1)\\ g_2^2(\bar{a}_2-1)\\ g_3^2(\bar{a}_3-1)\\ g_4^2(\bar{a}_4-1)\end{bmatrix}$$

Seed and prior

plane (two-point: $\bar{a}(1) = 0,\ \bar{a}'(1) = 1,\ \bar{a}''(1) = -\frac{327}{200},\ d_1 - c_1 = \frac{9}{8}$):

$(c_0, c_1, d_1, d_2) = \left(\frac{102}{127},\ -\frac{229}{127},\ -\frac{689}{1016},\ -\frac{1}{8}\right)$

cell (Fig. 11 band fit):

$(c_0, c_1, d_1, d_2) = (0.353,\ -0.571,\ 0.484,\ 0.058)$

(2026-09-16 contact-height coordinate $\bar{w}'_s = \bar{w}_s + 1:\ c'_0 = c_0 + c_1 + 1,\ c'_1 = c_1 + 2,\ d'_1 = d_1 + 2,\ d'_2 = d_2 + d$

$\hat{c}_0[0] = c_{0,\text{seed}}, \quad \hat{c}_1[0] = c_{1,\text{seed}}, \quad \hat{d}_1[0] = d_{1,\text{seed}}, \quad \hat{d}_2[0] = d_{2,\text{seed}}$

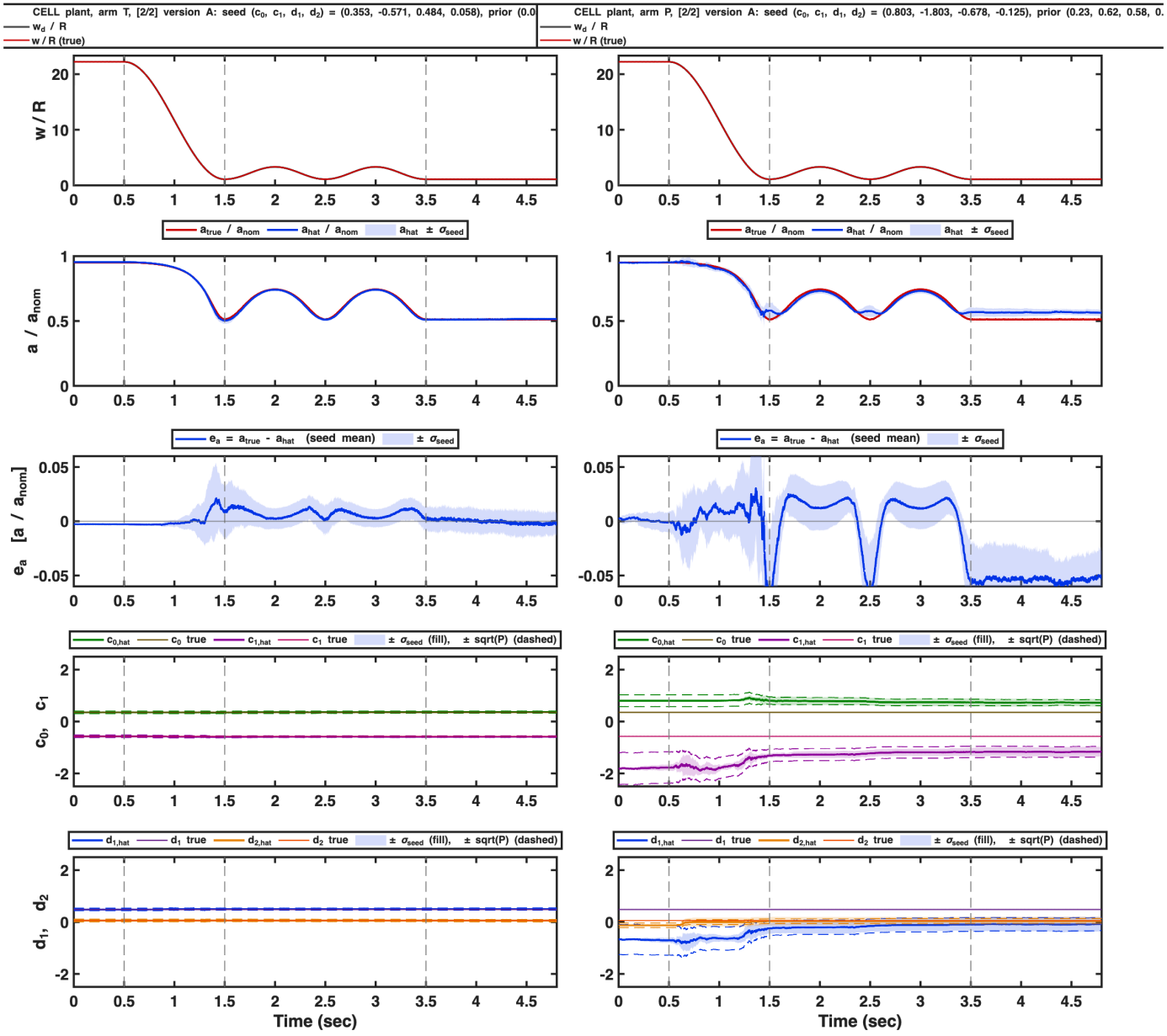
$$\hat{\hat{a}}_w[0] = \frac{N(\bar{w}[0] - \bar{w}_s;\ \hat{c}_0[0], \hat{c}_1[0])}{D(\bar{w}[0] - \bar{w}_s;\ \hat{d}_1[0], \hat{d}_2[0])}$$

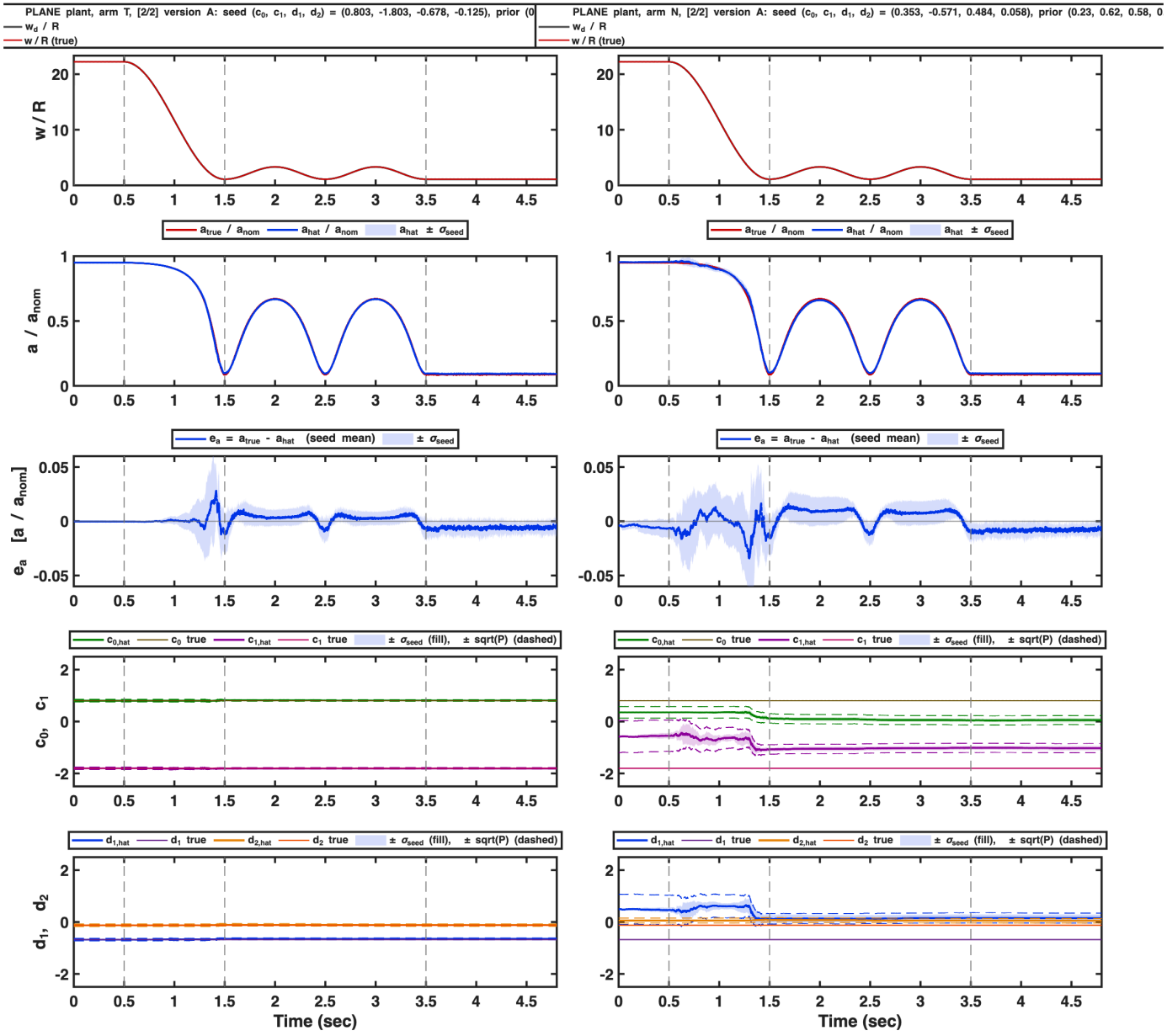
$$P[0] = \text{diag}\big(P_{11}[0],\ P_{22}[0],\ P_{33}[0],\ \sigma^2_{c_0,0},\ \sigma^2_{c_1,0},\ \sigma^2_{d_1,0},\ \sigma^2_{d_2,0}\big)$$

wall family (half the plane–cell spread, in this basis): $(\sigma_{c_0,0},\ \sigma_{c_1,0},\ \sigma_{d_1,0},\ \sigma_{d_2,0}) = (0.225,\ 0.616,\ 0.581,\ 0.0$

tight (own band fit): 0.05 on all four

denominator guard at the band floor: $D(1.1) \geq 2\, g_{\text{floor}}$


CELL plant, arm T, [2/2] version A: seed (c0, c1, d1, d2) = (0.353, -0.571, 0.484, 0.058), prior (0.0
CELL plant, arm P, [2/2] version A: seed (c0, c1, d1, d2) = (0.803, -1.803, -0.678, -0.125), prior (0.23, 0.62, 0.58, 0.
w_d / R
w / R (true)
w / R
a_true / a_nom
a_hat / a_nom
a_hat ± σ_seed
a / a_nom
e_a = a_true - a_hat (seed mean)
± σ_seed
e_a [a / a_nom]
c_0,hat
c_0 true
c_1,hat
c_1 true
± σ_seed (fill), ± sqrt(P) (dashed)
c0, c1
d_1,hat
d_1 true
d_2,hat
d_2 true
d1, d2
Time (sec)


PLANE plant, arm T, [2/2] version A: seed (c0, c1, d1, d2) = (0.803, -1.803, -0.678, -0.125), prior (0
wd / R
w / R (true)
PLANE plant, arm N, [2/2] version A: seed (c0, c1, d1, d2) = (0.353, -0.571, 0.484, 0.058), prior (0.23, 0.62, 0.58, 0
wd / R
w / R (true)
w / R
atrue / anom
ahat / anom
ahat ± σseed
a / anom
ea = atrue - ahat (seed mean)
ea [a / anom]
c0,hat
c0 true
c1,hat
c1 true
± σseed (fill), ± sqrt(P) (dashed)
c0, c1
d1,hat
d1 true
d2,hat
d2 true
d1, d2
Time (sec)